

ETIOLOGY AND PATHOGENESIS

Acute Cutaneous Graft-Versus-Host Reaction after Allogeneic Stem Cell Transplantation

The precise pathophysiology of GVHD is not known. A distillation of many lines of evidence implicates allostimulation of donor lymphocytes by major and/or minor histocompatibility antigens expressed on recipient tissues, leading to immune activation and tissue damage.

TABLE 28-1

Terms and Definitions

- **Allogeneic:** Between two individuals
- **Autologous:** From the same person
- **Syngeneic:** Between two individuals with identical genetic material
- **Xenograft:** Transplanted tissue from another species
- **Graft-versus-host disease (GVHD):** The totality of organ dysfunction attributable to the targeting of the recipient's tissues by immunocompetent lymphocytes
- **Graft-versus-host reaction (GVHR):** The expression of GVHD in a specific organ (e.g., cutaneous GVHR)

- **Stage:** A term referring to clinical findings
- **Grade:** A term referring to histopathologic findings

TABLE 28-2

Overview of Acute Graft-Versus-Host Disease (GVHD) by Type of Transplant

Transplant Type	Incidence and Severity of GVHD	Cutaneous Manifestations	Treatment	Histopathology
Allogeneic, matched	++	Erythematous macules and papules, variable distribution; bullae rare	Prednisone, FK-506, Anti-thymocyte globulin, Monoclonal antibodies	Dermal infiltrate of lymphocytes, more than four necrotic epidermal cells per linear millimeter, sub-epidermal bulla
Allogeneic, mismatched	+++	Generally greater distribution and intensity than for allogeneic, matched	Same	Same
Allogeneic, unmatched	+++++	Generally greater distribution and intensity than for allogeneic, matched	Same	Same
Allogeneic, unmatched, unrelated	+++++	Generally greater distribution and intensity than for allogeneic, matched	Same	Same
Syngeneic	+	Generally less widely distributed than for allogeneic, matched	Usually none	Fewer lymphocytes, fewer necrotic keratinocytes than for allogeneic, matched
Autologous	+	Generally less widely distributed than for allogeneic, matched	Usually none	Changes resemble eruption of lymphocyte recovery
Autologous with cyclosporine	++	Generally less widely distributed than for allogeneic, matched	Usually none	Fewer lymphocytes, fewer necrotic keratinocytes than for allogeneic, matched
Autologous with cyclosporine and interferon-γ	++	Eruption more widespread, erythroderma more common than for autologous with or without cyclosporine	Prednisone rarely for erythroderma	Spongiosis often a prominent feature with fewer necrotic epidermal cells than for allogeneic, matched

Note: + to +++++ = less common/severe to most common/severe.